

The axisymmetric velocity field corresponding to a three-dimensional steady vortex of circulation Γ is given in cylindrical coordinates by

$$\begin{aligned}v_r &= -\frac{r}{\tau_r}, \\v_\theta &= \frac{\Gamma}{2\pi r} \left\{ 1 - \exp \left[-\left(\frac{r}{R_o} \right)^2 \right] \right\} \\v_z &= \frac{z}{\tau_z},\end{aligned}$$

where the constants R_o , τ_r and τ_z measure the radius of the inner core where viscous effects are important, and the characteristic times associated with radial convection and axial stretch, respectively.

1. Determine the condition required for the motion to be that of an incompressible fluid.
2. Compute streamlines and trajectories.

$$\nabla \cdot \vec{v} = \frac{1}{r} \left[\frac{\partial}{\partial r} r v_r + \frac{\partial v_\theta}{\partial \theta} + r \frac{\partial v_z}{\partial z} \right] = \frac{1}{r} \left[-\frac{2r}{\tau_r} + \frac{r}{\tau_z} \right] = 0 \Rightarrow \boxed{\tau_r = 2\tau_z}$$

$$\frac{dr}{dt} = -\frac{r}{\tau_r} \quad \boxed{r = r_0 e^{-t/\tau_r}} \quad (T1)$$

$$\begin{aligned}r \frac{d\theta}{dt} &= \frac{\Gamma}{2\pi r} \left\{ 1 - e^{-\left(\frac{r}{R_o}\right)^2} \right\} \rightarrow r \frac{d\theta}{dr} = -\frac{\Gamma \tau_r}{2\pi r^2} \left\{ 1 - e^{-\left(\frac{r}{R_o}\right)^2} \right\} \\ \frac{dz}{dt} &= \frac{z}{\tau_z} \rightarrow \boxed{z = z_0 e^{t/\tau_z}} \quad (T3) \\ &\rightarrow \boxed{\theta - \theta_0 = -\frac{\Gamma \tau_r}{2\pi} \int_{r_0}^r \left[1 - e^{-\left(\frac{r}{R_o}\right)^2} \right] \frac{dr}{r^3}} \quad (T2)\end{aligned}$$

THE STREAMLINES ARE GIVEN BY (T2) AND $\left(\frac{z}{z_0}\right)^{\tau_z} = \left(\frac{r}{r_0}\right)^{-\tau_r} \Rightarrow \boxed{z r^2 = z_0 r_0^2}$